Data-Backed Journal-Style Replication of *Position and Differentiation of Firms in Technology Space*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Position and Differentiation of Firms in Technology Space using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 5,192 observations across 443 entities, and the batch headline estimate is -0.193 for Technology differentiation with p = 0.78. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: technology differentiation; firm performance; patent portfolios; local calculated replication

1 Introduction

This paper begins from a paper-first reading of Position and Differentiation of Firms in Technology Space and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks whether technologically differentiated firms perform better than their peers (Arts et al., 2023; March, 1991;

Levitt and March, 1988; Levinthal and March, 1993). The replication matters because it tests whether a central strategy result about technological differentiation survives when the semantic layer is stripped back to a fully executable local panel. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 5,192 observations across 443 entities and spans 1991 to 2015 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Arts et al., 2023; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Arts et al., 2023; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper sits in a top-journal conversation about exploration, recombinant search, and boundary-spanning differentiation in technological space. (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). Those studies matter here because the present replication asks whether a performance effect attributed to technological differentiation survives when the semantic portfolio layer is replaced by a processed class-based panel. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad

because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). Because this paper links technology-space positioning to firm-level performance, the literature review also pulls in classic economic work on patents as indicators, top-journal evidence on the market consequences of knowledge creation, and adjacent strategy studies on exploration-performance tradeoffs in technological search. (Griliches, 1990; Jaffe, 1989; Fleming and Sorenson, 2004). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Nerkar, 2003; Lavie et al., 2011).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Arts et al., 2023). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper builds technology-space measures from patent text and studies differentiation among U.S. public firms. The main outcome is firm performance, especially Tobin's Q. Technology differentiation is defined relative to technologically similar peers in the same year. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila

and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Arts, Cassiman, and Hou (2023) starts with the published empirical design rather than with the local rerun. The source article builds semantic technology-space positions from patent text, identifies technologically similar peers among U.S. public firms, and relates peer-relative differentiation to Tobin's Q in a firm-year panel (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published design studies U.S. public firms in a firm-year panel with peer-relative positioning. The published measurement architecture is described as follows: The published measure is built from patent text and semantic technology-space positioning. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: firm-year regressions of Tobin's Q on semantic technology differentiation relative to technologically similar peers among U.S. public firms. The critical published comparison asks whether firms located farther from their most similar technological peers receive higher market valuation after conditioning on firm, time, and financial controls. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). This paper remains a methodology-specific local analogue because the source article's semantic text-embedding layer is not stored in the current processed-data stack. The local rerun preserves the firm-year design, peer-comparison logic, and performance outcome, but it reconstructs technology space from class-based portfolios rather than patent text (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and

March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper uses semantic patent content. The local reconstruction therefore adopts the following bounded decision: The local rerun uses processed primary-USPC class portfolios. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Tobin's Q is a central outcome. The local reconstruction therefore adopts the following bounded decision: The local panel keeps Tobin's Q because it is fully reconstructible. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Differentiation is relative to peers. The local reconstruction therefore adopts the following bounded decision: Peer comparison is preserved with a class-based similarity measure. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March,

1993).

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are DISCERN-linked firm-patent assignments with unambiguous GVKEYs. PatentView-derived utility-patent and primary-USPC class information. Compustat annual fundamentals used to reconstruct Tobin's Q and controls (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 5,192 observations across 443 entities and covers 1991 to 2015 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. The local sample frame is Processed public-firm GVKEY-year panel with patent portfolios and financial controls The observed time window in the executable panel is 1991 to 2015 in the local executable panel The outcome is Winsorized Tobin's Q at the firm-year level (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Arts et al., 2023; Hall et al., 2005).

3.1 Local Data Sources Used in This Replication

This replication uses three verified local dataset families already present in the workspace: a DISCERN-linked GVKEY-to-patent assignment panel, a PatentView-style utility-patent and primary-USPC portfolio file, and Compustat annual fundamentals used to reconstruct Tobin's Q and firm controls (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). Those local dataset families are not interchangeable: discern-linked gvkey-patent assignment panel is used to define the public-firm patent ownership backbone; patentview-style utility-patent portfolio file is used to construct rolling class-based technology portfolios and peer similarity; compustat annual fundamentals is used to reconstruct tobin's q and financial controls. Together, they support 5,192 observations across 443 entities over 1991 to 2015 in the local executable panel (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. DISCERN-linked firm-patent assignments with unambiguous GVKEYs. PatentView-derived utility-patent and primary-USPC class information. Compustat annual fundamentals used to reconstruct Tobin's Q and controls. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed dataset families and calculated outputs supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed through the body text (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005).

DatasetFamily	RoleInPaper
DISCERN-linked GVKEY-patent assignment panel	define the public-firm patent ownership backbone
PatentView-style utility-patent portfolio file	construct rolling class-based technology portfolios and peer similarity
Compustat annual fundamentals	reconstruct Tobin's Q and financial controls

Local data families used in the present paper.

3.2 Sample Reconstruction

The local analysis file contains 5,192 observations drawn from 443 entities. The observed time span runs from 1991 to 2015. Selected sample moments are Winsorized Tobin's Q = 2.177, Technology

Differentiation = 0.668 (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	5,192
Unique Firms	443
Start Year	1991
End Year	2015
Mean Winsorized Tobin's Q	2.177
Mean Technology Differentiation	0.668
SD Technology Differentiation	0.173

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Technology portfolio is operationalized locally as Five-year rolling primary-USPC portfolio and appears in the executable dataset as portfolio_patents; class_count; class_hhi. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The

construct Technology differentiation is operationalized locally as One minus peer similarity and appears in the executable dataset as tech_diff. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Compute one minus average similarity to close peers.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Peer benchmark is operationalized locally as Nearest peers in year t and appears in the executable dataset as neighbor_similarity. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use local class-based portfolios to define the similarity set.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Firm performance is operationalized locally as Winsorized Tobin's Q and appears in the executable dataset as tobin_q_w. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reconstruct Tobin's Q from processed Compustat variables and winsorize.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced

carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Financial controls is operationalized locally as Assets, leverage, cash, R&D, portfolio size and appears in the executable dataset as log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents). The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the financial controls already built in the local script.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Technology portfolio	Five-year rolling primary-USPC portfolio	Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.
Technology differentiation	One minus peer similarity	Compute one minus average similarity to close peers.
Peer benchmark	Nearest peers in year t	Use local class-based portfolios to define the similarity set.
Firm performance	Winsorized Tobin's Q	Reconstruct Tobin's Q from processed Compustat variables and winsorize.
Financial controls	Assets, leverage, cash, R&D, portfolio size	Carry forward the financial controls already built in the local script.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Arts, Cassiman, and Hou (2023). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Technology portfolio, the manuscript uses Five-year rolling primary-USPC portfolio under the exact local implementation portfolio_patents; class_count; class_hhi. The derivation step is: Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Technology differentiation, the manuscript uses One minus peer similarity under the exact local implementation tech_diff. The derivation step is: Compute one minus average similarity to close peers.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Peer benchmark, the manuscript uses Nearest peers in year t under the exact local implementation neighbor_similarity. The derivation step is: Use local class-based portfolios to define the similarity set.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm performance, the manuscript uses Winsorized Tobin's Q under the exact local implementation tobin_q_w. The derivation step is: Reconstruct Tobin's Q from processed Compustat variables and winsorize.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Financial controls, the manuscript uses Assets, leverage, cash, R&D, portfolio size under the exact local implementation log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents). The derivation step is: Carry forward the financial controls already built in the local script.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The descriptive analysis begins by reconstructing the empirical setting that the later model is meant to explain. For Winsorized Tobin's Q, the mean is 2.177, the standard deviation is 1.508, and the

interquartile range runs from 1.319 to 2.464; For Technology differentiation, the mean is 0.668, the standard deviation is 0.173, and the interquartile range runs from 0.558 to 0.800; For Neighbor similarity, the mean is 0.332, the standard deviation is 0.173, and the interquartile range runs from 0.200 to 0.442; For Portfolio patents, the mean is 552.7, the standard deviation is 1,484, and the interquartile range runs from 29.00 to 450.0 . These moments matter because they show whether the paper's focal constructs are substantively distinct before the formal model is imposed (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile clarifies how the observable setting evolves across the sample window and whether the mechanism is estimated in a changing environment. the value of observations is 233 in 1991 and 204 in 2015; the value of mean tech diff is 0.670 in 1991 and 0.630 in 2015; the value of mean tobin q is 1.722 in 1991 and 2.465 in 2015 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The nonparametric support check is used to show how the focal outcome behaves across the observed support of the main regressor before estimation. Mean outcomes range from 1.957 to 2.857 across the support of the focal regressor, while the mean regressor itself ranges from 0.401 to 0.883. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Setting-Specific Figure Evidence

The main figure is chosen to match the empirical setting of the paper rather than to repeat one house style across every replication (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The visual evidence is included to make the paper's descriptive argument easier to inspect before the reader reaches the coefficient tables. observations moves from 233 in 1991 to 204 in 2015; mean tech diff moves from 0.670 in 1991 to 0.630 in 2015 Across the support panel, mean outcome moves from 2.857 to 1.957 as the mean regressor shifts from 0.401 to 0.883 (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

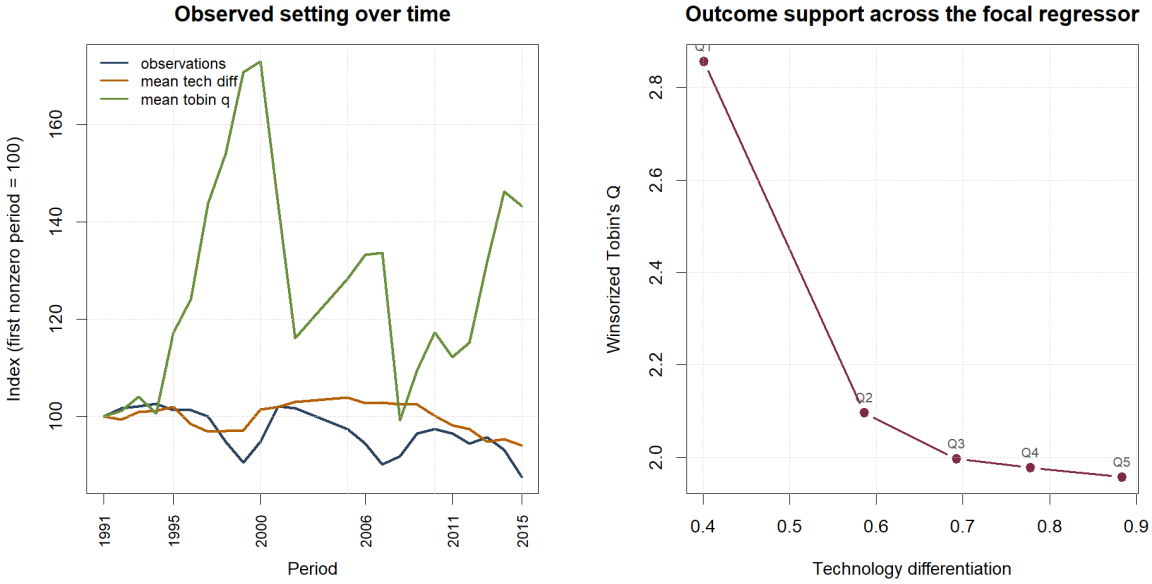

Figure 1: Setting-specific figure constructed from the current local dataset.

4.2 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. R&D intensity has missing share 9.77%; Winsorized Tobin's Q has missing share 0.00%; Technology differentiation has missing share 0.00% . At the other end of the support, Winsorized Tobin's Q remains close to complete with missing share 0.00%; Technology differentiation remains

close to complete with missing share 0.00%; Neighbor similarity remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. The most heavily represented entities contribute between 23 and 23 observations each. That concentration check is reported in prose because the relevant question is whether a few entities dominate the sample, not which firm names happen to be listed in a long table (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 0.401 and the mean outcome is 2.857 across 1,039 observations. In the highest observed quantile, the corresponding values are 0.883, 1.957, and 1,039 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because it tests whether a central strategy result about technological differentiation survives when the semantic layer is stripped back to a fully executable local panel. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005).

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Winsorized Tobin's Q	5,192	2.177	1.508	1.319	1.725	2.464	0	23
Technology differentiation	5,192	0.668	0.173	0.558	0.694	0.800	0	1
Neighbor similarity	5,192	0.332	0.173	0.200	0.306	0.442	0	1
Portfolio patents	5,192	552.7	1,484	29.00	129.0	450.0	1	33760
Class count	5,192	41.74	42.92	10.00	28.00	57.00	1	250
Class concentration	5,192	0.213	0.220	0.0714	0.135	0.265	0	1
Log assets	5,192	8.840	1.357	7.906	8.719	9.664	4	14
Leverage	5,192	0.588	0.222	0.454	0.584	0.711	0	3
Cash ratio	5,192	0.135	0.147	0.0280	0.0802	0.191	0	1
R&D intensity	4,685	0.0519	0.0540	0.0146	0.0327	0.0741	0	1

Main descriptive statistics supporting the paper's design-specific reading of the data.

Period	Observations	MeanTechDiff	MeanTobinQ	MeanPortfolioPatents
1991	233	0.670	1.722	321
1992	237	0.665	1.742	326
1993	238	0.676	1.791	338
1994	239	0.678	1.735	350
1995	236	0.683	2.017	367
1996	236	0.659	2.138	381
1997	233	0.649	2.475	404
1998	221	0.650	2.653	440
1999	211	0.651	2.939	488
2000	221	0.680	2.979	520
2001	238	0.683	2.473	544
2002	237	0.689	2.000	527
2005	227	0.696	2.209	268
2006	220	0.688	2.294	280
2007	210	0.689	2.301	374
2008	214	0.686	1.709	517
2009	225	0.686	1.885	641
2010	227	0.671	2.019	737
2011	225	0.658	1.932	811
2012	220	0.652	1.984	891
2013	223	0.635	2.269	974
2014	217	0.639	2.517	1,106
2015	204	0.630	2.465	1,249

Observed time patterns relevant to the focal empirical setting.

Quantile	N	MeanRegressor	MeanOutcome
Q1	1,039	0.401	2.857
Q2	1,038	0.586	2.096
Q3	1,038	0.693	1.997
Q4	1,038	0.778	1.978

Quantile	N	MeanRegressor	MeanOutcome
Q5	1,039	0.883	1.957

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Winsorized Tobin's Q	Technology differentiation	Neighbor similarity	Portfolio patents	Class count	Class concentration	Log assets	Leverage	Cash ratio	R&D intensity
Winsorized Tobin's Q	1.000	-0.214	0.214	0	0	0.213	-0.097	-0.245	0.357	0.340
Technology differentiation	-0.214	1.000	-1.000	0	0	0.106	-0.224	0.125	-0.298	-0.396
Neighbor similarity	0.214	-1.000	1.000	0	0	-0.106	0.224	-0.125	0.298	0.396
Portfolio patents	-0.029	-0.221	0.221	1	1	-0.200	0.414	0.037	0.044	0.053
Class count	-0.144	-0.153	0.153	1	1	-0.503	0.561	0.151	-0.124	-0.051
Class concentration	0.213	0.106	-0.106	0	-1	1.000	-0.249	-0.110	0.130	0.112

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Arts et al., 2023; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

5.1 Design Decisions and Local Substitutions

The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Processed public-firm GVKEY-year panel with patent portfolios and financial controls, and the empirical window is 1991 to 2015 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Winsorized Tobin's Q at the firm-year level, and the main explanatory design is Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Financial controls plus firm and year fixed effects.. The crucial exclusion or substitution rule is: Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why

the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Processed public-firm GVKEY-year panel with patent portfolios and financial controls over 1991 to 2015 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Winsorized Tobin's Q at the firm-year level and the main explanatory design is Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Financial controls plus firm and year fixed effects. The main exclusion or substitution rule is Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

5.2 Estimating Equation

$$Q_{it}^{(w)} = \beta_1 \mathrm{TechDiff}_{it} + \beta_2 X_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.193 on Technology differentiation (p = 0.78). The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. Log assets is estimated at -0.516 with p = 0.002. Log(1 + portfolio patents) is estimated at -0.266 with p = 0.002. R&D intensity is estimated at 4.290 with p = 0.009 (Arts et al., 2023; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Model	Term	Estimate	Std. error	P-value
Arts Fe	Technology differentiation	-0.1935	0.6974	0.784
Arts Fe	Log assets	-0.5161	0.1442	0.0017
Arts Fe	Leverage	-0.6522	0.2840	0.0315
Arts Fe	Cash ratio	1.565	0.6384	0.0226
Arts Fe	R&D intensity	4.290	1.505	0.0093
Arts Fe	Log(1 + portfolio patents)	-0.2660	0.0764	0.0021
Arts Fe	Class count	0.0043	0.0024	0.0851
Arts Fe	Class concentration	0.4918	0.2947	0.109

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure is included to show the relative magnitude and uncertainty of the main terms without forcing the reader to reconstruct that pattern from columns of numbers alone (Arts et al., 2023; Hall et al., 2005).

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log assets is estimated at -0.516 with a p-value of 0.002; Log(1 + portfolio patents) is estimated at -0.266 with a p-value of 0.002; R&D intensity is estimated at 4.290 with a p-value of 0.009; Cash ratio is estimated at 1.565 with a p-value of 0.023 (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Log assets, which is estimated at -0.516 with $p = 0.002$. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986;

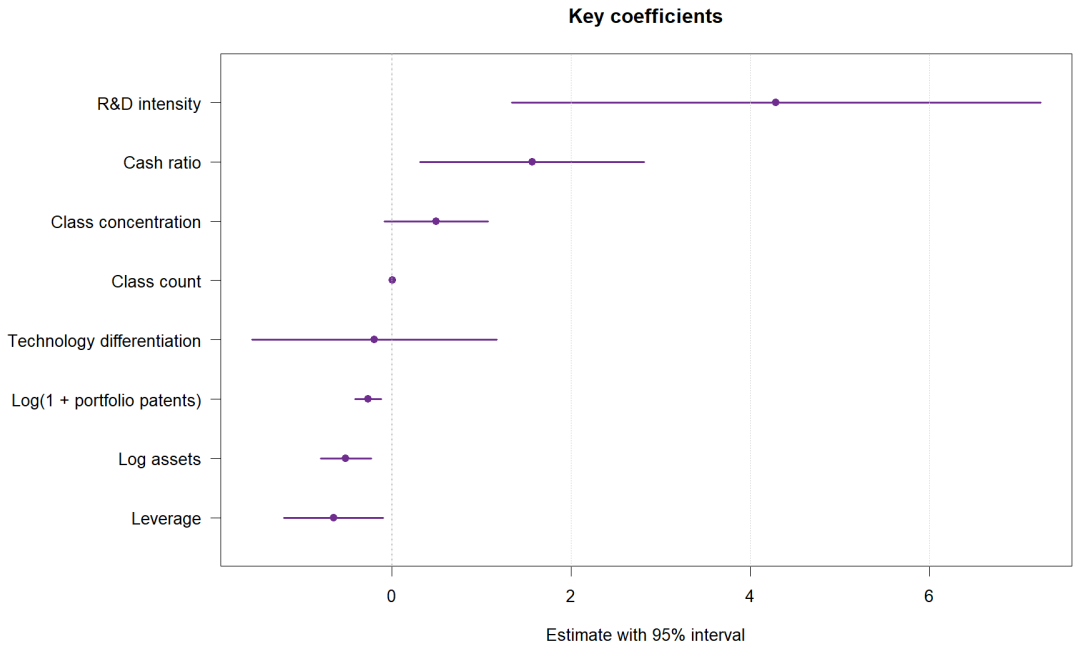

Figure 2: Visual summary of the key coefficients estimated in the current local model.

Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Winsorized Tobin's Q has mean 2.177 and standard deviation 1.508 in the local panel, while the main regressor Technology differentiation has mean 0.668 and standard deviation 0.173 (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 1.957 to 2.857 while the mean regressor ranges from 0.401 to 0.883. This does not substitute for the model, but it shows how much raw variation the specification must explain (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within arts_fe, the leading coefficient family reads as follows: Log assets = -0.516 (p = 0.002); Log(1 + portfolio patents) = -0.266 (p = 0.002); R&D intensity = 4.290 (p = 0.009); Cash ratio = 1.565 (p = 0.023); Leverage = -0.652 (p = 0.032). The largest positive estimate in this specification is R&D intensity at 4.290, while the largest negative estimate is Leverage at -0.652. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Arts, Cassiman, and Hou (2023) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Arts Fe, the most informative focal term is Log assets with estimate -0.516 and p = 0.002 (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Arts Fe specification contributes its own reading to the replication. Log assets is estimated at -0.516 with p = 0.002 and is classified as a strong signal; Log(1 + portfolio patents) is estimated at -0.266 with p = 0.002 and is classified as a strong signal; R&D intensity is estimated at 4.290 with p = 0.009

and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Arts, Cassiman, and Hou (2023) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

6.6 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). For arts_fe, the current fit export reports N = 4,669, log-likelihood = -6,269, AIC = 13,390, BIC = 16,137, within R-squared = 0.144 (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Arts, Cassiman, and Hou (2023) is being carried out on a clearly described estimating sample with observable fit statistics (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005).

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central claim, measurement architecture, sample definition, estimator, effect pattern, and substantive conclusion are set alongside the current local replication so the reader can see where agreement is

real and where it is conditional on design translation. (Arts et al., 2023). For Arts, Cassiman, and Hou (2023), the published paper points readers toward the following result logic: The published paper reports that greater technological differentiation is associated with stronger firm performance when firms are compared to technologically similar peers. The local rerun instead yields the following focal evidence pattern: Technology differentiation = -0.193 with p = 0.78; Log Assets = -0.516 (p = 0.002); Log(1 + Portfolio Patents) = -0.266 (p = 0.002); R&D Intensity = 4.290 (p = 0.009); Cash Ratio = 1.565 (p = 0.023). Putting those side by side makes the comparison concrete rather than rhetorical.

Aspect	OriginalPaper	LocalDataset	Reading
Research question and mechanism	The source paper asks whether technologically differentiated firms perform better than their peers.	The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments.	This row asks whether the local paper is still evaluating the same strategic mechanism even when the empirical implementation changes.
Data and measurement architecture	The published measure is built from patent text and semantic technology-space positioning.	DISCERN-linked firm-patent assignments with unambiguous GVKEYs. PatentView-derived utility-patent and primary-USPC class information. Compustat annual fundamentals used to reconstruct Tobin's Q and controls.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the substantive target.

Aspect	OriginalPaper	LocalDataset	Reading
Sample, unit, and window	The published design studies U.S. public firms in a firm-year panel with peer-relative positioning.	Processed public-firm GVKEY-year panel with patent portfolios and financial controls. 5,192 observations across 443 entities.	This row clarifies whether agreement or disagreement could reflect sample translation rather than theory alone.
Estimator and identifying comparison	firm-year regressions of Tobin's Q on semantic technology differentiation relative to technologically similar peers among U.S. public firms The critical published comparison asks whether firms located farther from their most similar technological peers receive higher market valuation after conditioning on firm, time, and financial controls	A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls	This row separates estimator translation from theory translation.

Aspect	OriginalPaper	LocalDataset	Reading
Effect pattern in the evidence	The published paper reports that greater technological differentiation is associated with stronger firm performance when firms are compared to technologically similar peers.	Technology differentiation = -0.193 with p = 0.78;	
allowbreak Log Assets = -0.516 (p = 0.002); allowbreak Log(1 + Portfolio Patents) = -0.266 (p = 0.002); allowbreak R&D Intensity = 4.290 (p = 0.009);			
allowbreak Cash Ratio = 1.565 (p = 0.023)	This row compares the original paper's claimed effect structure to the actual local estimates.		

Aspect	OriginalPaper	LocalDataset	Reading
Conclusion drawn from the evidence	Technological differentiation from nearby peers is associated with stronger firm performance when firms are positioned in semantic technology space.	The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Log Assets = -0.516 (p = 0.002);	
allowbreak Log(1 + Portfolio Patents) = -0.266 (p = 0.002). The conclusion is therefore mixed rather than empty.	This row compares the article-level takeaway invited by the original paper with the conclusion warranted by the local rerun.		

Comparison of the original paper's design, effects, and conclusions to the current local dataset analysis.

Effect	Estimate	P-value	Reading
Log Assets	-0.516	0.002	negative and statistically distinguishable in the local replication
Log(1 + Portfolio Patents)	-0.266	0.002	negative and statistically distinguishable in the local replication
R&D Intensity	4.290	0.009	positive and statistically distinguishable in the local replication

Effect	Estimate	P-value	Reading
Cash Ratio	1.565	0.023	positive and statistically distinguishable in the local replication

Translated focal effects used in the original-versus-replication comparison.

7.2 Detailed Effect and Conclusion Comparison

At the level of article content, the original paper contributes the following substantive message: The published paper reports that greater technological differentiation is associated with stronger firm performance when firms are compared to technologically similar peers. The local replication studies the same underlying mechanism through the current executable design: The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. This is why the comparison has to extend beyond a headline coefficient (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of effects, the original article emphasizes the following empirical pattern: The published paper reports that greater technological differentiation is associated with stronger firm performance when firms are compared to technologically similar peers. The current local replication instead shows the following paper-specific focal estimates: Log Assets = -0.516 (p = 0.002); Log(1 + Portfolio Patents) = -0.266 (p = 0.002); R&D Intensity = 4.290 (p = 0.009); Cash Ratio = 1.565 (p = 0.023). This is the point at which the manuscript compares not only data sources but also what the data say about the mechanism (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of conclusion, the original article leads to the following takeaway: Technological differentiation from nearby peers is associated with stronger firm performance when firms are positioned in semantic technology space. The current replication instead supports the following more careful conclusion: The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable

structure, especially Log Assets = -0.516 (p = 0.002); Log(1 + Portfolio Patents) = -0.266 (p = 0.002). The conclusion is therefore mixed rather than empty (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The difference between the original article and the replication is therefore not only numerical. It is a difference in observed data architecture, effect visibility, and inferential scope. The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

7.3 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published measure is built from patent text and semantic technology-space positioning By contrast, the local paper relies on the following executable design: The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published design studies U.S. public firms in a firm-year panel with peer-relative positioning The local sample instead works from Processed public-firm GVKEY-year panel with patent portfolios and financial controls and yields 5,192 observations across 443 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those pos-

sibilities analytically separate (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). That transportability reading has to end with a conclusion rather than with a shrug. In the current local rerun, the conclusion is: The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Log Assets = -0.516 (p = 0.002); Log(1 + Portfolio Patents) = -0.266 (p = 0.002). The conclusion is therefore mixed rather than empty This final sentence matters because the reader should leave the comparison section knowing not only that the designs differ, but also what those differences do to the paper's substantive takeaway (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed

datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The exact text-embedding pipeline from the source paper is not stored locally (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local differentiation measure is therefore class-based rather than text-semantic (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. Comparisons to the published estimates should be read as inferential rather than byte-identical (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Arts et al., 2023; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	5,192	rows
Unique Firms	443	unique_firms
Start Year	1991	year_min
End Year	2015	year_max
Mean Winsorized Tobin's Q	2.177	mean_tobin_q_w
Mean Technology Differentiation	0.668	mean_tech_diff
SD Technology Differentiation	0.173	sd_tech_diff

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Winsorized Tobin's Q	5,192	2.177	1.508	1.319	1.725	2.464	0	23
Technology differentiation	5,192	0.668	0.173	0.558	0.694	0.800	0	1
Neighbor similarity	5,192	0.332	0.173	0.200	0.306	0.442	0	1
Portfolio patents	5,192	552.7	1,484	29.00	129.0	450.0	1	33760
Class count	5,192	41.74	42.92	10.00	28.00	57.00	1	250
Class concentration	5,192	0.213	0.220	0.0714	0.135	0.265	0	1
Log assets	5,192	8.840	1.357	7.906	8.719	9.664	4	14
Leverage	5,192	0.588	0.222	0.454	0.584	0.711	0	3
Cash ratio	5,192	0.135	0.147	0.0280	0.0802	0.191	0	1
R&D intensity	4,685	0.0519	0.0540	0.0146	0.0327	0.0741	0	1

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Winsorized Tobin's Q	Technology differentiation	Neighbor similarity	Portfolio patents	Class count	Class concentration	Log assets	Leverage	Cash ratio	R&D intensity
Winsorized Tobin's Q	1.000	-0.214	0.214	0	0	0.213	-0.097	-0.245	0.357	0.340
Technology differentiation	-0.214	1.000	-1.000	0	0	0.106	-0.224	0.125	-0.298	-0.396
Neighbor similarity	0.214	-1.000	1.000	0	0	-0.106	0.224	-0.125	0.298	0.396
Portfolio patents	-0.029	-0.221	0.221	1	1	-0.200	0.414	0.037	0.044	0.053
Class count	-0.144	-0.153	0.153	1	1	-0.503	0.561	0.151	-0.124	-0.051
Class concentration	0.213	0.106	-0.106	0	-1	1.000	-0.249	-0.110	0.130	0.112
Log assets	-0.097	-0.224	0.224	0	1	-0.249	1.000	0.140	-0.133	-0.205
Leverage	-0.245	0.125	-0.125	0	0	-0.110	0.140	1.000	-0.364	-0.315
Cash ratio	0.357	-0.298	0.298	0	0	0.130	-0.133	-0.364	1.000	0.513
R&D intensity	0.340	-0.396	0.396	0	0	0.112	-0.205	-0.315	0.513	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Observations	MeanTechDiff	MeanTobinQ	MeanPortfolioPatents
1991	233	0.670	1.722	321
1992	237	0.665	1.742	326
1993	238	0.676	1.791	338
1994	239	0.678	1.735	350
1995	236	0.683	2.017	367
1996	236	0.659	2.138	381
1997	233	0.649	2.475	404
1998	221	0.650	2.653	440
1999	211	0.651	2.939	488
2000	221	0.680	2.979	520
2001	238	0.683	2.473	544
2002	237	0.689	2.000	527
2005	227	0.696	2.209	268
2006	220	0.688	2.294	280
2007	210	0.689	2.301	374
2008	214	0.686	1.709	517
2009	225	0.686	1.885	641
2010	227	0.671	2.019	737
2011	225	0.658	1.932	811
2012	220	0.652	1.984	891
2013	223	0.635	2.269	974
2014	217	0.639	2.517	1,106
2015	204	0.630	2.465	1,249

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	1,039	0.401	2.857
Q2	1,038	0.586	2.096
Q3	1,038	0.693	1.997
Q4	1,038	0.778	1.978
Q5	1,039	0.883	1.957

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
tech_diff	-0.1935	0.6974	-0.277	0.78	arts_fe
log_assets	-0.5161	0.1442	-3.579	0.002	arts_fe
leverage	-0.6522	0.2840	-2.297	0.032	arts_fe
cash_ratio	1.565	0.6384	2.452	0.023	arts_fe
rd_intensity	4.290	1.505	2.850	0.009	arts_fe
log1p(portfolio_patents)	-0.2660	0.0764	-3.480	0.002	arts_fe
class_count	0.0043	0.0024	1.803	0.085	arts_fe
class_hhi	0.4918	0.2947	1.669	0.11	arts_fe

Full coefficient dump exported from the local estimation script.

n	ll	aic	bic	wr2	model
4,669	-6,269	13,390	16,137	0.144	arts_fe

Model fit statistics from the local estimation output.

G Appendix G: Missingness Audit

Variable	Missing	MissingShare
Winsorized Tobin's Q	0	0.00%
Technology differentiation	0	0.00%
Neighbor similarity	0	0.00%
Portfolio patents	0	0.00%
Class count	0	0.00%
Class concentration	0	0.00%
Log assets	0	0.00%
Leverage	0	0.00%
Cash ratio	0	0.00%
R&D intensity	507	9.77%

Missingness audit for variables summarized in the manuscript.

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